

The Case of an Unconventional CSRF Attack in Firefox

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Content

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It appears that an unconventional method of [Cross Site Request Forgery](#)) may be made exploitable by using Firefox versions 21 and below. The exploit requires that the target application be first vulnerable to HEAD request [verb tampering](#), which is where a HEAD verb(also commonly known as 'method') is supplied in place of a GET or POST, and is successfully processed by the application. Once this is found, an [XMLHttpRequest](#)(commonly abbreviated to 'XHR') request can be sent from an off-domain location with the [.open\(\)](#) method invoked and HEAD supplied as the verb.

The XMLHttpRequest Living Standard specifications can be found [here](#) and defines how XHR objects should be used. Although there are many rules, steps 3 and 4 of the [.send\(\)](#) method serve particular interest to this implementation error:

.send(data);

1. If the request method is GET or HEAD, set data to null.
1. If data is null, do not include a request entity body and go to the next step.

Consider the following very basic and elementary Proof of Concept:

```
<script>
var xhr = new XMLHttpRequest();
var url = "https://www.whitehatsec.com";
var data = "foo=bar";
xhr.withCredentials = true; //Allows for sending cookies with the request
xhr.open("HEAD", url, true);
xhr.setRequestHeader("Content-type", "application/x-www-form-urlencoded");
xhr.send(data);
</script>
```

If you monitor your traffic or catch this in an intercepting proxy, you will see a request being made to <https://www.whitehatsec.com> with post data "foo=bar", even though the request verb is HEAD. According to step 3 above, 'data' should have been set to 'NULL'. This behavior seems to only occur in Firefox; The latest versions(as of this writing) of Internet Explorer, Chrome, Safari, and Opera are all successfully practicing proper .send() implementation.

I [notified](#) Mozilla of this behavior and a patch has been implemented into version 22 that was released on June 25th, 2013, and only users using a previous version of Firefox would be vulnerable to this now. It requires a bit of a "perfect storm" scenario, but could be extremely damaging depending on the context of the vulnerable application. I'd like to extend a huge thanks to the Mozilla security team for the swift attention this received, as well as for allowing me to participate in the remediation process.

[Mozilla Foundation Security Advisory 2013-54](#)

[Mitre CVE-2013-1692](#)